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## Anatomy of sovereign yield behaviour using textual news

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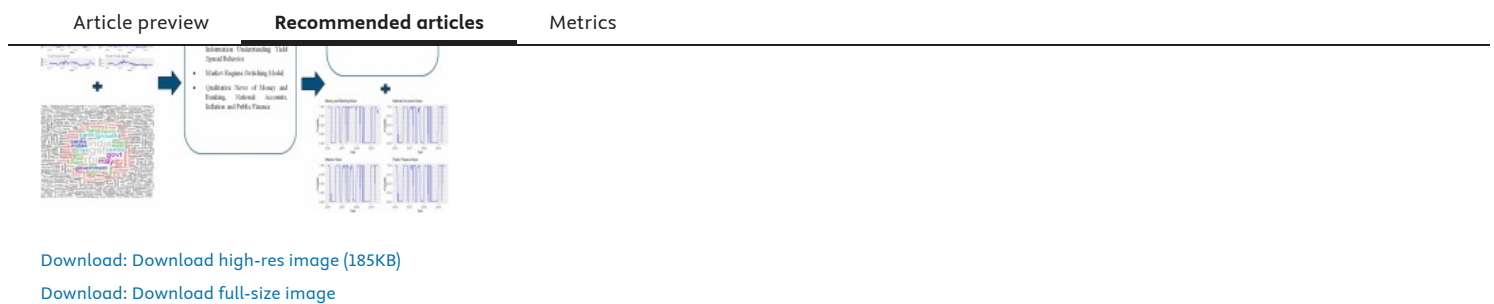
### Highlights

- We examine the impact of textual news on sovereign bond yield spreads.
- Sentiment scores come from bond market news published in newspapers.
- The articles were bifurcated into positive and negative news.
- Textual news sentiment explains both steepening and flattening of the yield curve.
- Monetary and fiscal policy news have the most significant impact.

### Abstract

While the relationship between the information content of macroeconomic news and the behavior of asset prices has been studied extensively in the finance literature, this study provides a new perspective by examining the impact of textual news on sovereign bond yield spreads in an emerging country. This study used bond market news published in newspapers to develop the sentiment scores using a modified word dictionary to unravel news characteristics. A nonlinear regime-shifting regression model of Markov Regime Shifting (MRS) is used to understand the impact of news on sovereign bond yield spreads. The paper results show that textual news sentiment may explain both steepening and flattening of the yield curve, with monetary and fiscal policy news having the most significant impact on yield spread behaviour. The results hold key implications for policymakers, debt fund managers and other market participants.

### Graphical Abstract



## JEL classification

G11; G14; G15; G19; G20

## Keywords

yield spread; sovereign bond; textual news; macroeconomic news

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## Introduction

The relationship between the information content of macroeconomic news and the behavior of asset prices has been studied extensively in the finance literature. The literature attempts to decode the impact of macroeconomic news arrival and their release timing on the volatility in asset prices (Andersson et al., 2006, Corbet et al., 2020, El Ouadghiri et al., 2016, Elder et al., 2012, Hautsch et al., 2011, Jones et al., 1998, Maslyuk-Escobedo et al., 2017, Nikkinen et al., 2006, Rangel, 2011, Shen et al., 2022, Simpson and Ramchander, 2004, Valseth, 2013, Zaremba et al., 2020). Although prior empirical research focused on quantitative news announcements, recent literature attempted to explore newer frontiers, such as analyzing textual analysis using natural language processing, which has served as a good proxy for market sentiment and demystify asset price dynamics (Twedt and Rees, 2012, Bartov et al., 2018). Textual news content can serve as a private information proxy intrinsic to testing the efficient market hypothesis (Shiller, 2000) and also reflects the brewing market sentiment before the release of quantitative information (Banerjee et al., 2021).

The present paper broadens the literature by providing new insights by examining the impact of textual news on the term structure behavior of yield spreads using different tenors of the Indian sovereign bonds. We used the broad categories of macroeconomic news using the extant empirical literature but generated their sentiment scores using text-mining methods.

Significant studies examine the responses to quantitative news releases on asset price formation (McNeil and Frey, 2000, Campbell et al., 2001, Elder et al., 2012, Banerjee and Pradhan, 2021, Boitan and Marchewka-Bartkowiak, 2022). The studies on sovereign bond markets primarily have focused on capturing the effect of macroeconomic data releases on the level of yields (Poterba and Rueben, 1999; Özatay et al., 2009, Beetsma et al., 2013, Kim et al., 2015, Caporale et al., 2018). For instance, Goldberg and Leonard (2003) and Jaramillo and Weber (2013) established that inflation and employment-related news releases, such as consumer price inflation and nonfarm payrolls, significantly affect sovereign bond yields (Gürkaynak et al., 2010). Özatay et al. (2009), Wright (2011), and Bauer et al. (2014) reported that the term structure of bond yields is sensitive to uncertainty in inflation (Duffee, 2018).

| Article preview                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Recommended articles | Metrics |
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| <p>bond yields beyond macroeconomic fundamentals, such as liquidity risk and risk aversion (Remolona et al., 2008, Alquist, 2010, Caceres et al., 2010, Afonso et al., 2012, Bai et al., 2012, Jaramillo and Weber, 2013 Boudt and Petitjean, 2014). In the context of sovereign bond markets, the impact of news arrivals on the shape and shift of yields are few. For example, Ramchander et al. (2005) and Altavilla et al. (2017) explored the link between uncertainty in macroeconomic fundamentals and the treasury bond yield spread. Likewise, Goldberg and Grisse (2013) reported time variation in the bond yield spreads following US Federal Reserve announcements.</p> <p>Nayak (2010) studied the influence of investor sentiment on corporate bond yield spreads. The study results indicated that sentiments impact bond yields and drive mispricing in the bond market, which aligns with the stock market. The study by Spyrou (2013) supports the findings of Nayak (2010), stating that investor sentiments are a key determinant of changes in bond yields. In contrast, Karlsson and Österholm (2020) explored the link between the corporate bond yield spread and the real economy. Using a Bayesian VAR model, they reported that corporate bond yield spread aids in forecasting real economic activity that tends to continue over the period (see also Azad et al., 2018). Cepni et al. (2020) studied the forecasting abilities of a newer investor sentiment index to predict the above-normal return of the US government-issued shorter-period debt instruments. The study results show that the newer index has a greater ability to predict the associated bond premia for bonds with shorter maturities (see also Pham and Cepni, 2022).</p> <p>Studying the factors influencing sovereign bond yield spreads, Bernoth and Erdogan (2012) posited that government debt levels, changing risk perception, and investor aversion significantly affect interest differentials, whereas Eichler (2014) found that political determinants profoundly affect yield spread behavior, especially with countries practicing autocracy and closed regimes against democracy and openness (see also Bernoth et al., 2012). Like Bernoth and Erdogan (2012), Afonso and Kazemi (2022) examined the co-movements and spillovers among EMU countries. Using generalized methods of moments (GMM) and VAR in a panel framework, they reported that lag spreads of Italian and Austrian bonds negatively affect the yield spreads of the rest of the countries under study while that of Irish, Portuguese, Belgian, and French raised the overall spread. Silvapulle et al. (2016) reported that the Euro Interbank Offered Rate, stock index returns, volatility index (VIX), and sovereign debt ratings influenced the bond yield spreads. Further, financial contagion was overwhelming among the peripheral EU countries of Germany, Portugal, Italy, Ireland, Greece and Spain (Caporin et al., 2018, Codogno et al., 2003, Costantini et al., 2014, Maltritz, 2012).</p> <p>The current stream of literature has investigated the impact of unstructured text news accessible through print media, online repositories, and social media platforms to check complementary scales on asset pricing (Garcia et al., 2003; Bartov et al., 2018, Banerjee et al., 2021, Cepni et al., 2022a, Cepni et al., 2023a, Akyildirim et al., 2022, Cepni and Khorunzhina, 2023). These studies establish that analyzing unstructured textual information helps uncover the significance of intrinsic sentiments much more robustly than its quantitative counterpart (see also Cepni et al., 2022b; Cepni et al., 2023b). Extant literature that used textual news, such as public filing information with the US Securities and Exchange Commission (SEC), corporate disclosures, and social media platforms like Twitter (Tetlock, 2007, Tetlock et al., 2008, Loughran and McDonald, 2011, Twedt and Rees, 2012, Bartov et al., 2018, Aysan et al., 2024) mainly focussed on equities or the cryptocurrency market (Akyildirim et al., 2021, Akyildirim et al., 2023, Cepni and Aysan, 2023).</p> <p>Banerjee et al. (2021) found evidence of textual news impacting the volatility of Indian sovereign benchmark bonds using high-frequency data. Here again, the method of textual news analysis is undertaken primarily in the context of bond yields, and the studies dealing with yield spreads are rather limited (See also De Santis, 2020). For instance, Gotthelf and Uhl</p> |                      |         |

Ireland, Italy, Portugal, and Spain) (see also Von Hagen et al., 2011). Dergiades et al. (2015) found that the volume of activity on social media and Google search queries increased Irish and Greek sovereign bond yield spreads.

The present study focuses on the Indian sovereign bond market, which has USD 2.3 trillion in outstanding government securities. The presence of the Indian bonds as a part of the JP Morgan Chase Government Bond Index – Emerging Markets further reflects the growing relevance of the Indian bond market, primarily sovereign-issued instruments.<sup>1</sup>

To our knowledge, the paper explores the role of textual news sentiment on the behavior of sovereign bond yield spreads in an emerging market like India. The paper uses a wide range of news categories for fiscal, monetary, and national accounts that are key drivers of yield spreads, drawing from the term structure literature. Our study helps us better understand how unexpected news impacts the yield spreads of sovereign bonds over various tenures. It helps to demystify any atypical characteristics embodied in the shape and shifts in term structure. The paper attempts to extract investor sentiment from soft or qualitative information from daily financial media, which gauges investor sentiment.

We use the Markov regime-switching regression (MRS) model to capture temporal asymmetries and nonlinear dynamics. The regime-switching approach enables our analysis to reflect changes in policy regimes across the study period. Unlike Gotthelf and Uhl (2019), who envisaged a linear relationship, we anticipate the relationship to be nonlinear. Hence, we use a nonlinear technique regime-shifting model following Hamilton, 1996, Hamilton, 2011.

Our results show that regime shifts in the sovereign bond yield spread to most news sentiments. We found the impact of monetary, fiscal, and national account news on yield spreads over tenors. Our results support the findings of Sibbertsen et al. (2014). Our conclusions hold practical implications for policymakers exploring the shifts and shape of sovereign bond term structures. These findings also provide useful information on portfolio rebalancing for debt fund managers on tenor-based spread reactions to surprises in the news.

This paper proceeds as follows. Section 2 presents the data and descriptive statistics. Section 3 provides the methodology. Section 4 reports the empirical results, and Section 5 sets our conclusions.

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## Section snippets

### Data and descriptive statistics

The source of textual news data is from leading Indian financial newspapers, which are the source of information signals (see for signaling theory Banerjee and Pradhan, 2020, Banerjee et al., 2020). We used formal commercial media rather than social media for the current study. The data sourced in such a way tends to be more robust given the expert base of

## MRS model

Our study uses the MRS model to capture temporal asymmetries and nonlinear dynamics. Before estimating the MRS model, we estimated a baseline linear regression model where the sovereign bond yield spread is regressed against the explanatory variables as follows:

$$YS_{i,t} = \alpha_{0,t} + \beta_{i,t} X_{i,t} + \varepsilon_{i,t}$$

where  $YS_{i,t}$  represents the yield spread, i.e., the difference in yield between two different tenured bonds,  $\alpha_{0,t}$  is the constant,  $X_{i,t}$  is the vector of explanatory variables and  $\varepsilon_{i,t}$  is the independent and ...

## Descriptive statistics

Table 2 presents descriptive statistics of the sovereign yield spread series, credit risk, liquidity risk, and risk aversion measures. All series have positive mean values over the sample period. In contrast, both the yield spread and control variables are distinctly non-Gaussian and are negatively or positively skewed with excess kurtosis, thus indicating leptokurtic distribution. As expected, the spread between 15–1-year is highest, and this gradually narrows over 15–3-year, 15–5-year, and ...

## Conclusion

A growing body of literature studies the role of textual news sentiment in equity and cryptocurrency markets; few have examined the influence of soft or qualitative news in bond markets, especially regarding spread behavior. This paper analyses the impact of textual news sentiment on the sovereign debt markets in an emerging economy (India). It examines the role of qualitative information on bond yield spreads in news reports. The period covered by this research is from January 01, 2016, to ...

## CRedit authorship contribution statement

**Susan Dann:** Writing – original draft, Visualization, Supervision. **Ahmet Sensoy:** Writing – original draft, Visualization, Supervision, Investigation, Conceptualization. **Md Akhtaruzzaman:** Writing – original draft, Visualization, Validation, Conceptualization. **H K Pradhan:** Writing – original draft, Software, Methodology, Formal analysis, Data curation. **Ameet Kumar Banerjee:** Writing – original draft, Methodology, Formal analysis, Data curation. ...

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